Course Project: Secure Communication Design and Implementation

Overview

In this project, students are required to **design and implement a secure communication scheme** between two untrusted parties. The objective is to apply concepts covered in the course to ensure key security properties such as **confidentiality, integrity, authentication, and other relevant features**.

Students are strongly encouraged to align this project with their **Final Year Project (FYP)**, where applicable, by securing communication between components or endpoints within their system.

Objectives

- Apply cryptographic and security principles to real-world systems
- Design secure communication protocols ensuring **CIA (Confidentiality, Integrity, Availability)**
- Incorporate additional properties such as authentication and non-repudiation
- Justify the choice of security mechanisms used

Initial Submission (Project Titles)

Each group must:

- Form groups of **up to four (4) students**
- Propose a suitable application scenario
- Submit the following details via the provided **Google Sheet**:
    - Project title
    - Group member names and registration numbers
    - https://docs.google.com/spreadsheets/d/1x16XHfh0RF6FnI7H_FImlexgDYBsfN7YD-HKfh6B7R0/edit?usp=sharing

📅 **Deadline:** *30th April*

Project Requirements

Students must:

- ***Design and Implement*** a communication mechanism between **two untrusted entities**

- Use appropriate techniques (e.g., encryption, hashing, key exchange, authentication protocols)
- Consider potential threats and how the design mitigates them
- Justify all design choices based on security requirements

Final Evaluation

The completed projects will be evaluated during the week starting from:

📅 **18th May**

Evaluation components:

- **Presentation**
- **Live Demonstration (if applicable)**
- **Viva (oral examination)**

Students are expected to:

- Clearly explain their system and design decisions
- Demonstrate how security properties are achieved
- Defend their approach against potential attacks